TOPIC DBMS Class Assignment : 01 DATE 26/02/26

Name - Utsah Roll no. - 2401010203
Course - B. Tech CSE (Sec. A)

Ques1 ⇒ What are the key reasons why a DBMS is preferable to a traditional file system? How does it ensure data security and consistency?

Ques2 → A hospital wants to store patient, doctor and appointment data in a DBMS.
Identify the entities, attributes and relationships in the system. Draw the ER diagram and explain the key design decisions you made.

Ques3 ⇒ An online shopping platform needs to store data about customers, orders and products.
Suggest primary keys, foreign keys and integrity constraints for the system. How do these constraints help maintain the accuracy of the data?

Ques4 ⇒ Why is understanding Entity-Relationship models important when designing a database? What could happen if a relationship is not correctly defined between entities?

Answers ⇒ P.T.O (Page-3)

Teacher's Signature.....

TOPIC ..... DATE .....

Ans→1 Reasons for why DBMS is preferred–

a) Reduced data redundancy.
b) Data consistency.
c) Data integrity
d) DBMS provides automatic recovery after crashes.
e) It manages multiple users accessing data at the same time.

It ensures data security and consistency by–

- Authentication
- Authorization
- Encryption
- Integrity constraints (prevents invalid data entry)

Ans→2 Entities and attributes:

I) Patient
- → Patient_ID
- → Name
- → Age

II) Doctor
- → Doctor_ID
- → Name
- → Specialization

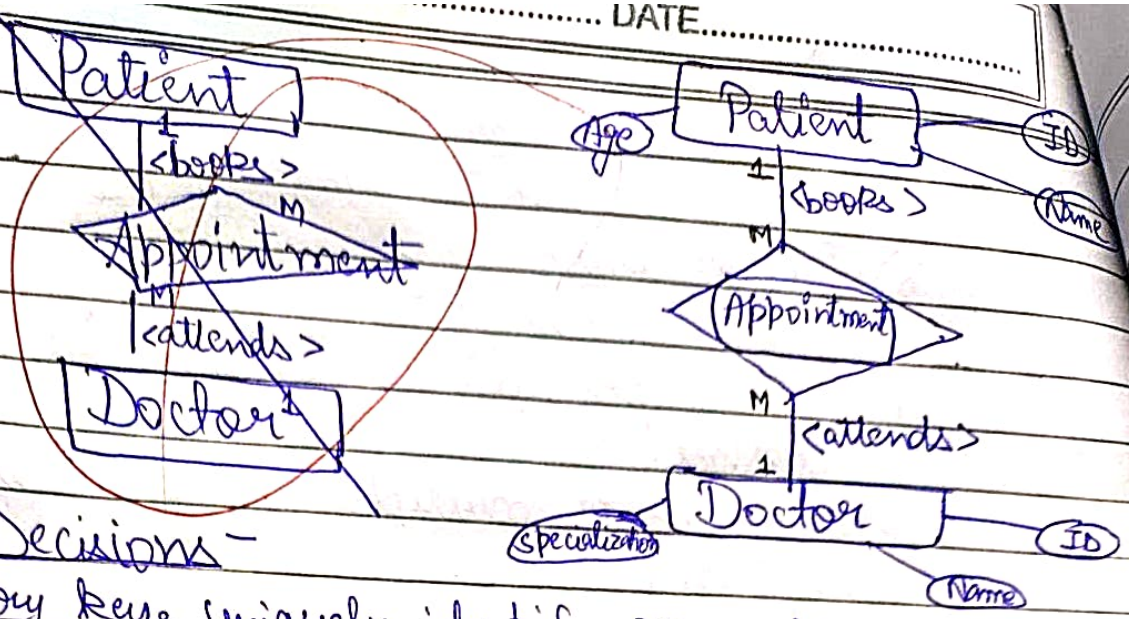
III) Appointment
- → Appointment ID
- → Date & time
- → Patient ID
- → Doctor ID

Relationships

> Patient can book many appointments.
> Doctor can have many appointments
> Appointments connect doctors and patients

Teacher's Signature.....

DATE.....



Key-Decisions-

- Primary keys uniquely identify records.
- Foreign keys maintain referential integrity.
- Avoided redundancy by separating entities.

Q=>3 Entities:

I) Customer
- ID
- Name
- Phone

II) Product
- ID
- Name
- Price

III) Order
- Order ID
- Order_date
- Customer ID

IV) Order details
- Order ID
- Product ID
- Quantity

DATE.....

Primary Key:
> uniquely identify each record.

Foreign key
> Maintains relationship between tables.
Example - Customer ID in order.
- Product ID in order_details.

Integrity constraints
1 > Entity integrity (Primary key cannot be null
> Referential integrity
> Domain const.
• Price > 0
• Stock ≥ 0
4 > Check constraints
Quantity > 0

Constraints maintain accuracy by :-
a) prevent duplicate entries.
b) prevent invalid references
c) prevent negative price & stock
d) Ensure valid data.

Q4 ⇒ ER models are important because-
• Helps in visualizing database structure
• Improve database efficiency
• Clearly defines entities and relationships
• Prevents redundancy
• Makes development easier.

TOPIC ..... DATE .....

What happens if relationships are not correctly defined –

1. data inconsistency
2. Integrity issues
3. Update anomalies
4. Deletion anomalies
5. Poor Query performance

Example –

If order and product relationship is not properly defined:

→ Same product may be repeated multiple times.

→ Deleting order might delete product details accidently.